pfi

Python for Fixed Income

COURSE 7 · NLP, RAG, AND LLMS

Clustering 10-K Risk Factors

WEEK 2 · DAY 3

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From search to grouping

Day 6: a question, then a search

The question chose what came back: semantic search over an indenture.

Today: no question, no label

K-means groups the risk factors; the course describes each group in the filers' own words.

18

risk factors, Boyd Gaming 10-K

48

risk factors, Levi Strauss 10-K

33

risk factors, Lamar 10-K

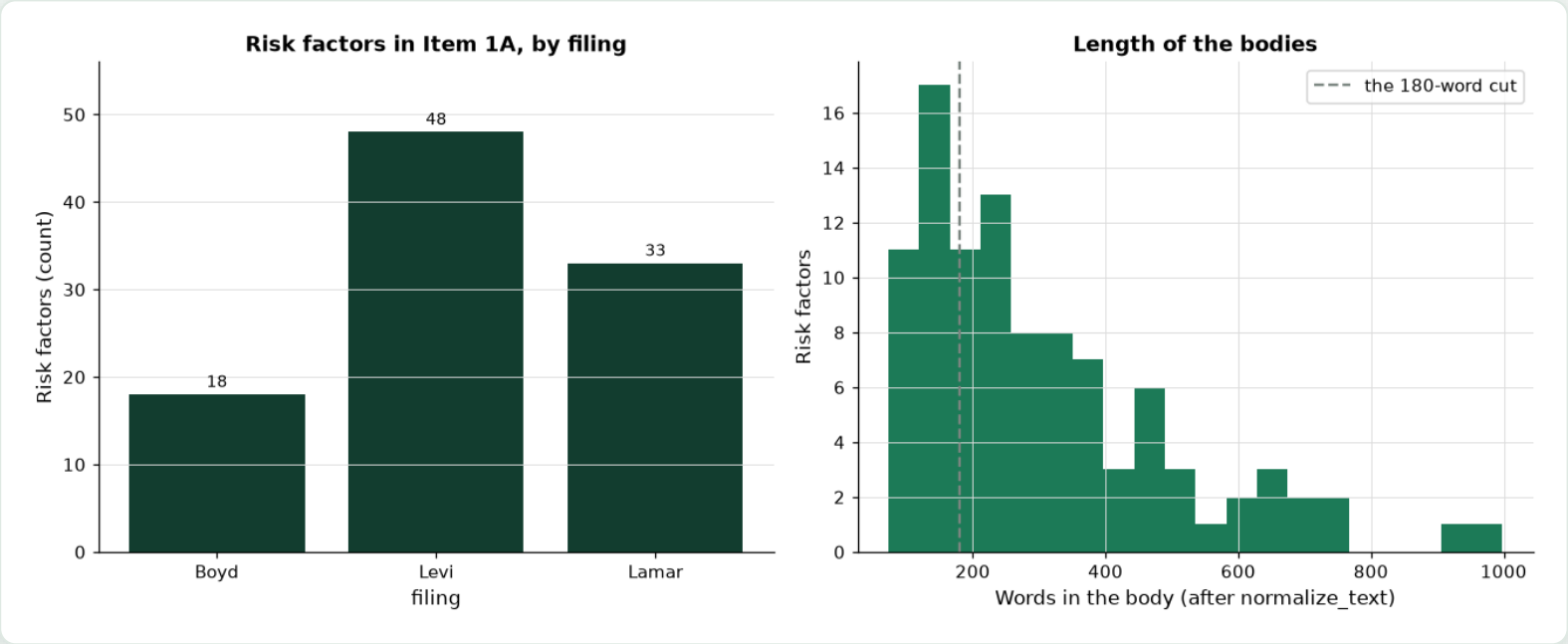
Clusters, described

A cluster of 10-K risk factors: its number, its terms, its counts per filing

Allowed names	Never called
a cluster	a risk ranking of the issuers
the risk factors in cluster 3	a measure of exposure
count of risk factors per filing	a credit view on a filer

A count of paragraphs is not a measure of risk: it shows how a filer drafts Item 1A.

Risk factors, counted per filing



99 risk factors; a body runs 73 to 998 words, median 241; 67 are longer than 180.

The counts describe the documents. They never rank the filers.

Source: Item 1A of the three Form 10-K filings, fiscal 2025, course7_risk_factors.csv.

One embedding per risk factor

```
texts = [heading + " " + first 180 words of the body]   # normalized
matrix = textkit.embed(model, texts)                  # once; every row of length 1
```

244

word pieces, the median text

26

of 99 texts over 256 pieces

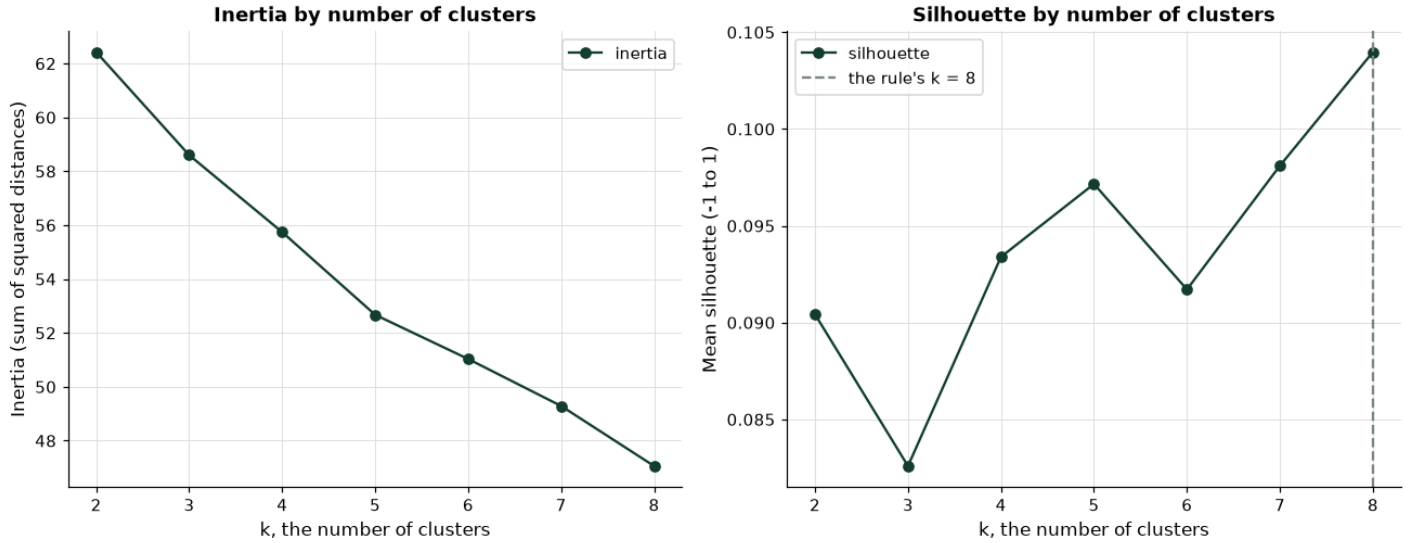
1.4481

squared distance = 2 - 2 x cosine

On unit-length rows, K-means groups by direction: the cosine of day 6.

K-means, and the rule for k

The rule, stated first: k from 2 to 8, the highest silhouette.



8

clusters, by the rule

0.1040

silhouette: clusters overlap

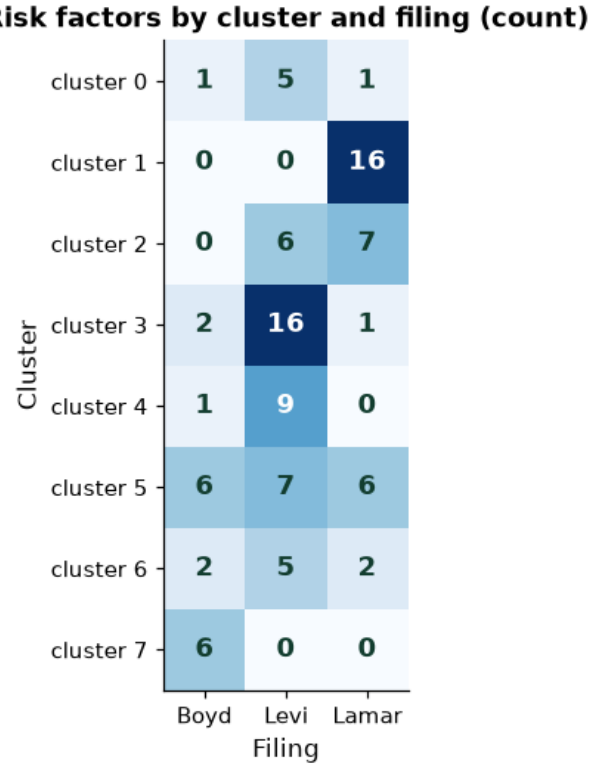
Every silhouette is near 0.1: a way to sort the risk factors for reading, not groups that stand far apart.

Describing a cluster: its terms

Cluster	Size	First five terms (the filers' words)
0	7	class, stock, common, securities, price
1	16	lamar, reit, income, advertising, taxable
2	13	outdoor, advertising, products, marketing, displays
3	19	retail, natural, distribution, stores, customers
4	10	laws, sanctions, countries, regulations, trade
5	19	debt, credit, facility, cash, indebtedness
6	9	information, data, systems, security, personal
7	6	gaming, casino, casinos, facilities, local

Lift: a term's mean TF-IDF inside the cluster minus outside it. Called by number, never named.

Clusters by filing, as counts

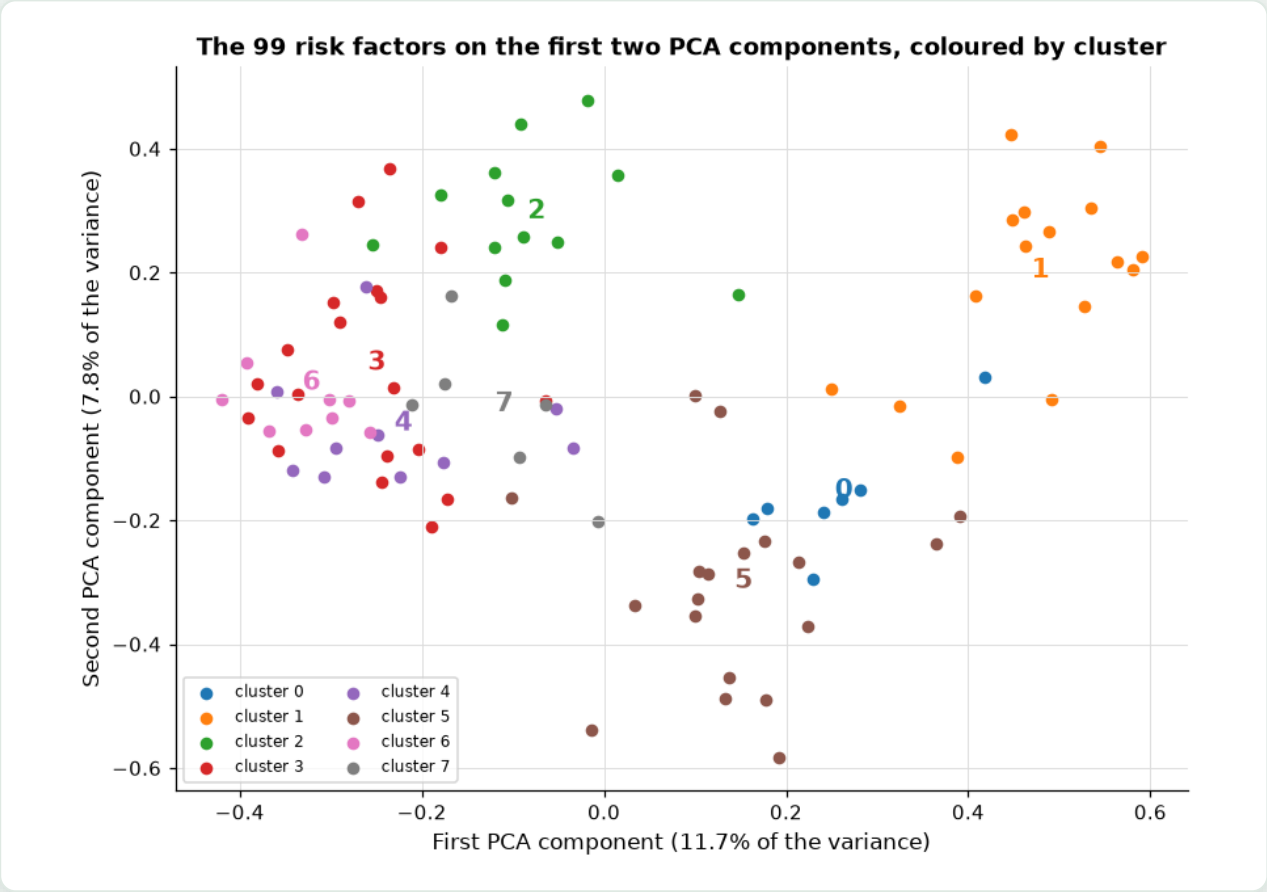


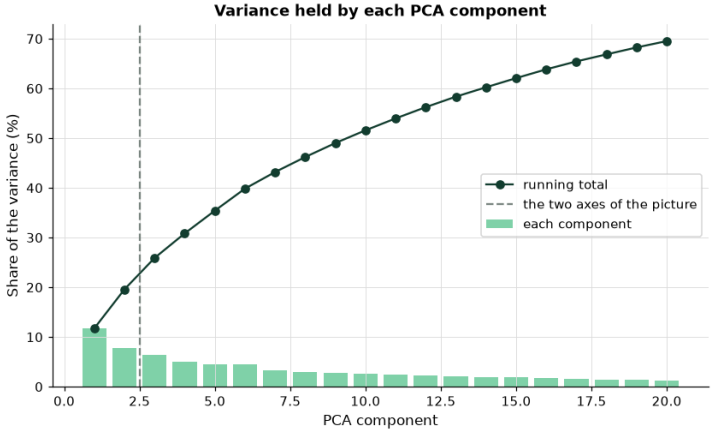
Cluster	Boyd	Levi	Lamar	All
0	1	5	1	7
1	0	0	16	16
2	0	6	7	13
3	2	16	1	19
4	1	9	0	10
5	6	7	6	19
6	2	5	2	9
7	6	0	0	6

A count of paragraphs is not a measure of risk.

The clusters move with the model, the 180-word cut, and k.

A picture in two dimensions, and what it hides





0.1958

of the variance on the two axes

0.8042

left off the picture

10 components hold half the spread, 48 hold 90 percent. A sketch, never a test.

Excel side by side

Excel side by side (tested in Excel)

`=COUNTIFS(risk_factors!$A$2:$A$100,B$1,risk_factors!$C$2:$C$100,$A2)`	grid equal: True
`=COUNTIFS(risk_factors!A2:A100,"Levi Strauss & Co.",risk_factors!C2:C100,3)`	16
`=COUNTIF(risk_factors!$A$2:$A$100,B$1)`	18, 48, 33
`=SUM(B2:D9)`	99

No sentence model or K-means in Excel: the clusters come from Python, the counting is COUNTIFS.

Save the day: one file's patch

```
git log -p -1 -- textkit.py
+def cluster_terms(texts: list[str], labels, n: int = 8) -> pd.DataFrame:
```

28 added lines, nothing removed. pytest: 42 passed.

The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.

1

Docstring

Units, conventions, and what the function must never do: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Assert what the docstring promises. Run it: it fails.

→

4

Fix

Find the rule it broke, fix it, run the test again.

Today: the piece vectors are summed, never scaled to length 1: rows 1.0000 to 5.0102, clusters by length.

Code that runs is not the same as code that is right.

What you can do now

- Count risk factors per filing, with source lines
- Embed once; unit rows make distance follow cosine
- Choose k by a rule stated first
- Describe a cluster by its terms, never a name
- Counts by filing: crosstab and COUNTIFS

Week 2, Day 4 Reading EDGAR